

L11 : Support Vector machines : Hard margin classifier

Agenda :

- Hard margin binary classifier
- Formulation as constrained convex optimization
- KKT conditions & dual formulation

So far, we ~~for~~ have focused on regression. We now turn to classification.

Recall: Classification is the brand of supervised learning wherein the output variable takes a discrete set of values; i.e. the output variable is qualitative / categorical.

Special case: Binary classification. Here, the o/p variable takes 2 values, they may be "coded" as $\{+1, -1\}$ or $\{0, 1\}$.

Eg: Does the input tissue smear contain cancer cells?

Is this the image of a man or a woman?

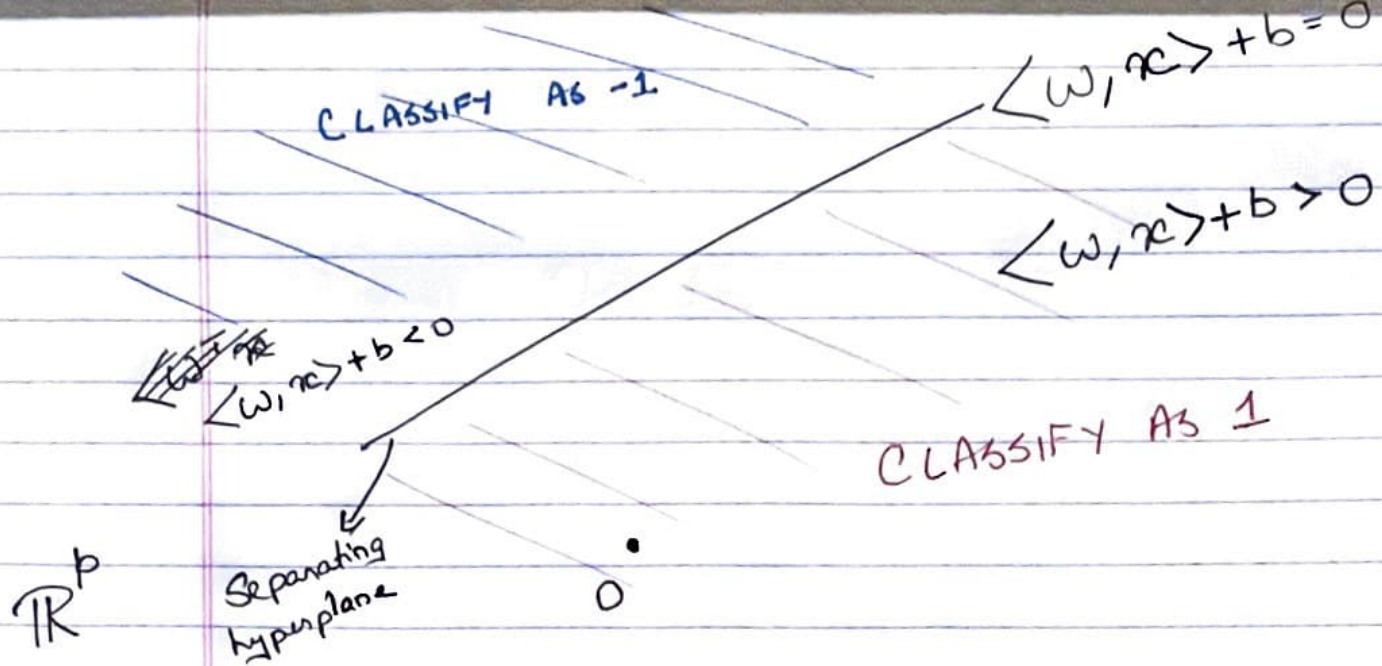
Today, we begin discussion of Support vector machines (SVM), which is a (set of) ^{optimisation based} approaches for binary classification.

Specifically, we code the outputs as $\{+1, -1\}$, and seek a classifier of the form

$$\begin{aligned} f(x) &= \text{sign}(w^T x + b) \\ &= \text{sign}(\langle w, x \rangle + b). \end{aligned}$$

Here, ~~the~~ the classification boundary is the "separating" hyperplane

$$\langle w, x \rangle + b = 0.$$



Goal: Compute (w, b) based on training data

Q: How?

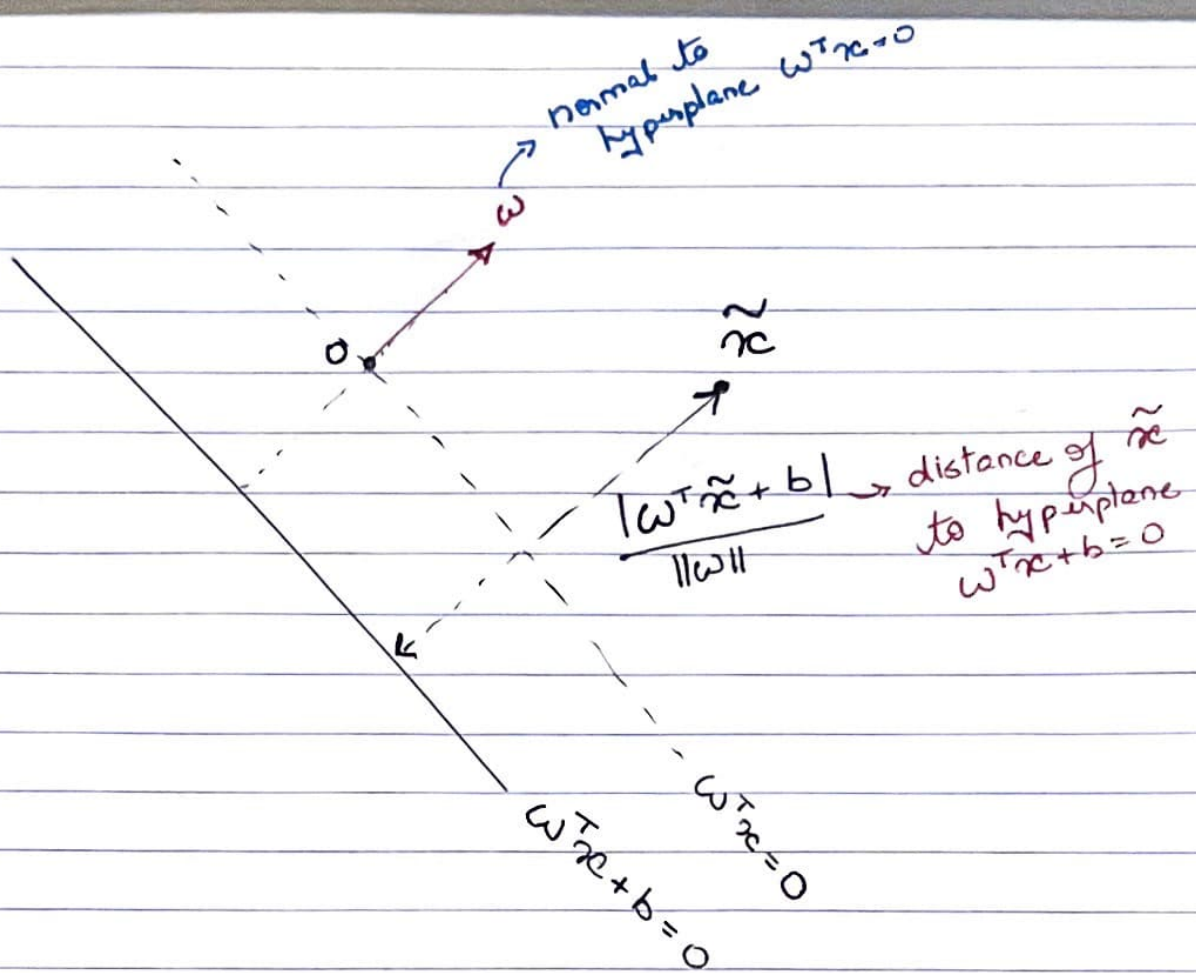
A: Find the separating hyperplane that "best" separates the training data.
 { Show examples }

Case 1: Training data is separable: Max margin classifier.

Assume first that there exists a separating hyperplane that separates the training data exactly (this will be relaxed later).

There exists (w, b) s.t.

$$y_i (w^T x_i + b) > 0 \quad \forall i$$



Def : The (geometric) margin of a separating hyperplane $w^T x + b = 0$ is:

$$p_{wb} = \min_{1 \leq i \leq n} \frac{|w^T x_i + b|}{\|w\|}$$

Captures distance of hyperplane to "nearest" training sample.

Max-margin classifier:

This is the ^{separating} hyperplane with the maximum margin (safest / most robust choice).

The max-margin classifier solves the optimization:

$$\max_{w, b: (w^T x_i + b) y_i > 0 \forall i} \min_{1 \leq i \leq n} \frac{|w^T x_i + b|}{\|w\|}$$

⇕

①

$$\max_{w, b} \min_{1 \leq i \leq n} \frac{y_i (w^T x_i + b)}{\|w\|}$$

{ INNER min is only +ve for separating hyperplanes! }

Note: $\frac{y_i (w^T x_i + b)}{\|w\|}$ is insensitive to

multiplicative ~~scat~~ scaling of (w, b) by +ve scalar.
 $\Rightarrow$ Without loss of generality (WLOG), we can
impose $\min_{1 \leq i \leq n} y_i (w^T x_i + b) = 1.$

This turns ① into:

$$\max_{w, b: \min_{1 \leq i \leq n} y_i (w^T x_i + b) = 1} \frac{1}{\|w\|}$$

⇕

②

$$\max_{(w, b): y_i (w^T x_i + b) \geq 1 \forall i} \frac{1}{\|w\|}$$

★ Solution of ② will satisfy
 $\min_{1 \leq i \leq n} y_i (w^*{}^T x_i + b^*) = 1$

Finally, we rewrite (2) in the following form:

Quadratic objective

$$\begin{array}{ll} \min_{w, b} & \frac{1}{2} \|w\|^2 \\ \text{s.t.} & y_i (w^T x_i + b) \geq 1 \quad \text{for } 1 \leq i \leq n \end{array}$$

MAX MARGIN CLASSIFIER
(constrained convex optimization)

n linear constraints

of (primal) optimization variable: $p+1$

of (primal) constraints: n

Next, let us explore what insights we can gather about the above optimization from its KKT conditions & dual.

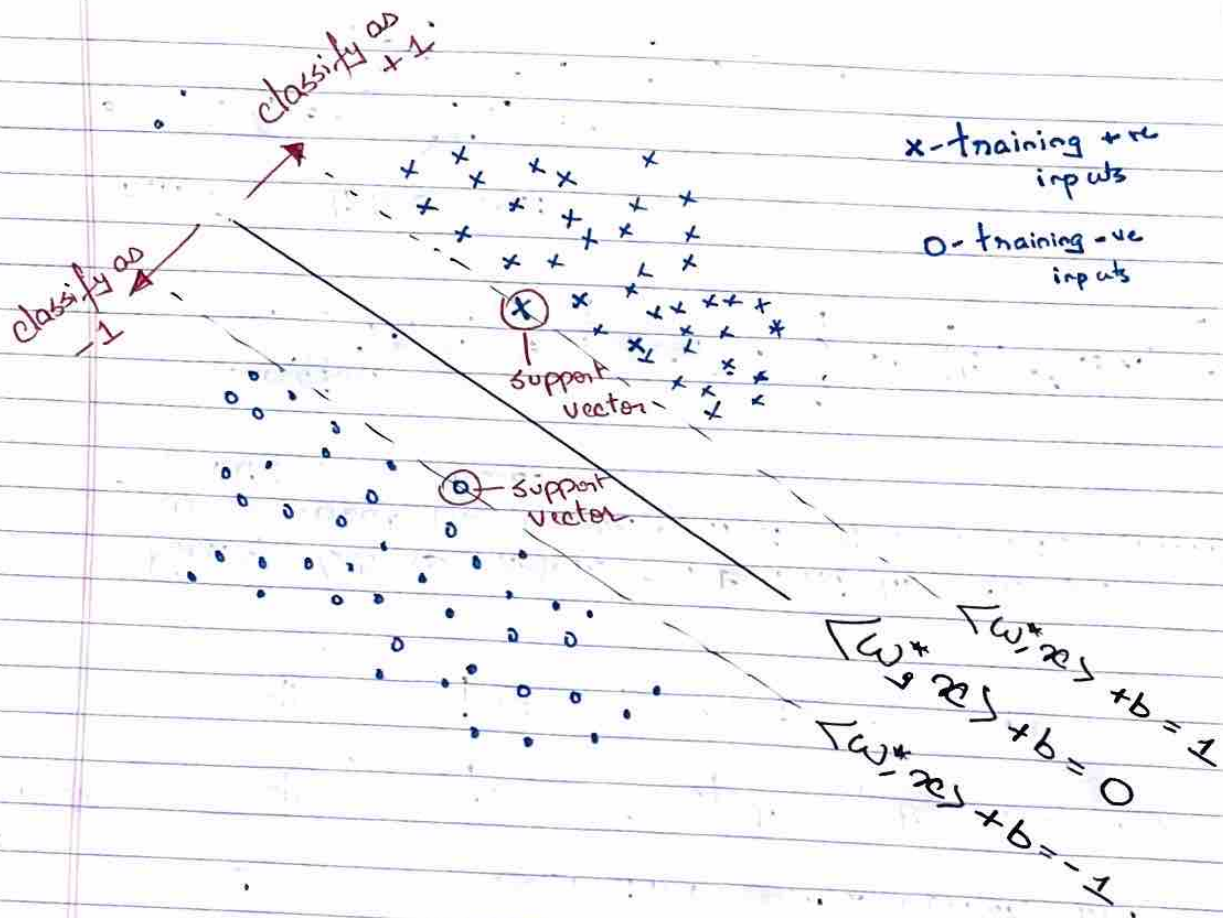
Lagrangian

$$\mathcal{L}(w, b, \alpha) = \frac{1}{2} \|w\|^2 - \sum_i \alpha_i [y_i (w^T x_i + b) - 1]$$

KKT Conditions are:

- $\nabla_w \mathcal{L} = 0, \quad \frac{\partial \mathcal{L}}{\partial b} = 0$

$$\nabla_w \mathcal{L} = w - \sum \alpha_i y_i x_i$$



$$\Rightarrow \boxed{\omega^* = \sum \alpha_i^* y_i x_i} \quad \text{--- (1)}$$

$$\nabla_b \mathcal{L} = - \sum_i \alpha_i y_i$$

$$\Rightarrow \boxed{\sum \alpha_i^* y_i = 0} \quad \text{--- (2)}$$

- Primal feasibility:

$$\boxed{y_i (\omega^{*T} x_i + b^*) \geq 1 \quad \forall i}$$

↳ (3)

- Complementary slackness:

$$\boxed{\alpha_i^* [y_i (\omega^{*T} x_i + b^*) - 1] = 0 \quad \forall i}$$

↳ (4)

- (4) $\Rightarrow$ The only +ve values of α_i^* correspond to those training inputs that are closest to the ~~sep~~ separation boundary - these are called ~~sep~~ support vectors.

From (1), it now follows ω^* is a linear combination of ~~only~~ the support vectors

★ Training inputs ~~not on the~~ that are not support vectors DO NOT influence the solⁿ.

Dual formulation

Recall:

$$\mathcal{L}(\omega, b, \alpha) = \frac{1}{2} \|\omega\|^2 - \sum \alpha_i [y_i(\omega^T x_i + b) - 1]$$

$$F(\alpha) = \inf_{\omega, b} \mathcal{L}(\omega, b, \alpha)$$

Note: Since $\mathcal{L}$ is linear in b , the inf above is $-\infty$ unless $\sum \alpha_i y_i = 0$.

$\Rightarrow$ We take $\sum \alpha_i y_i = 0$ as a constraint on the dual (suffices to maximize under this constraint!).

Under this constraint,

$$F(\alpha) = \sum_1^n \alpha_i - \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$$

{ Obtained by substituting $\omega = \sum \alpha_i y_i x_i$ into the Lagrangian }

$\Rightarrow$

Dual:

$$\begin{aligned} \max_{\alpha \in \mathbb{R}_+^n} \quad & \sum_{i=1}^n \alpha_i - \sum_{i=1}^n \sum_{j=1}^n \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle \\ \text{s.t.} \quad & \sum y_i \alpha_i = 0 \end{aligned}$$

Note: • The Dual objective is Concave (a quadratic convex function is fact).

- Can be solved by any convex. opt. solver.

Solving the dual & obtaining α_i^*

- Those i s.t. $\alpha_i^* > 0$ are the support vectors

- $\omega^* = \sum \alpha_i^* y_i x_i$

↳ ω^* is a linear combination of support vectors

- Fix a support vector x_i .

$$y_i [\omega^{*T} x_i + b^*] = 1$$

$$\Rightarrow b^* = \frac{-1}{y_i} - \omega^{*T} x_i$$

$$b^* = -\frac{1}{y_i} - \omega^{*T} x_i$$

- Final classifier is:

$$f^*(x) = \text{sign}(\langle \omega^*, x \rangle + b^*)$$

$$= \text{sign}\left(\sum_i \alpha_i^* y_i \langle x_i, x \rangle + b^*\right)$$

Primal
soln via
dual

★ Only need to compute dot product of test input x with the support vectors!